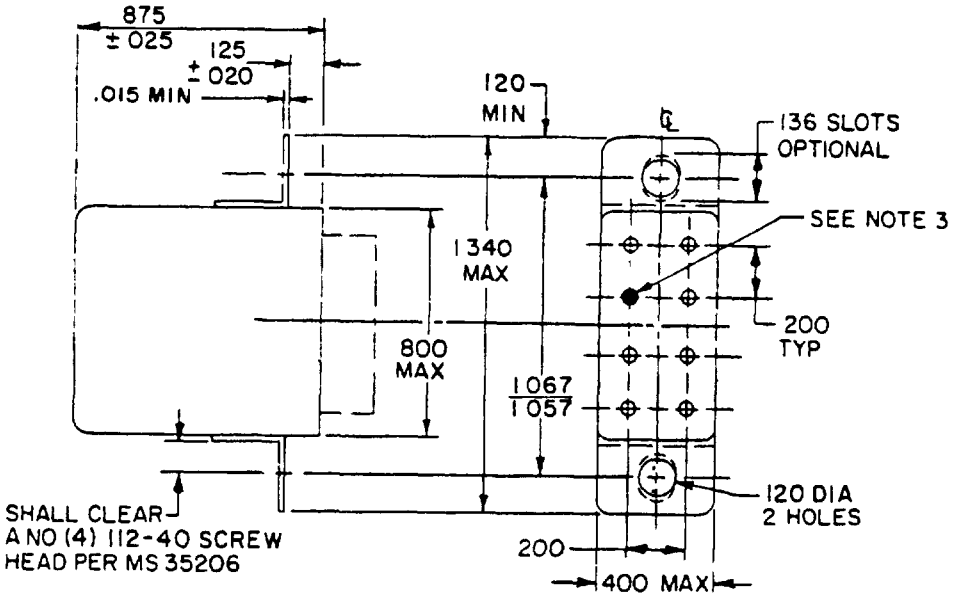
MILITARY SPECIFICATION SHEET

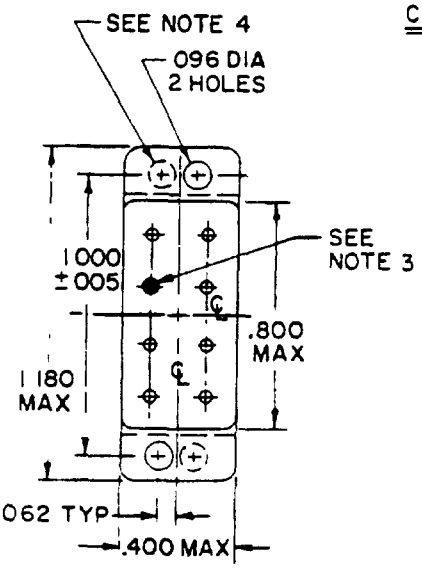
(K) RELAYS, ELECTROMAGNETIC, HERMETICALLY SEALED, DPDT, LOW LEVEL AND 2 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-5757.

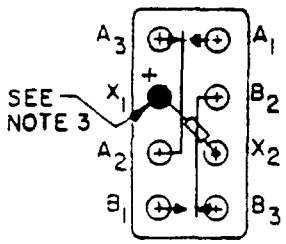



CONFIGURATION "A"




CONFIGURATION "B"

(INACTIVE FOR NEW DESIGN)




CIRCUIT DIAGRAM
TERMINAL VIEW

CONFIGURATIONS A, B, F, G, H AND J

DEENERGIZED POSITION

FIGURE 1 Dimensions and configuration

(K) denotes changes



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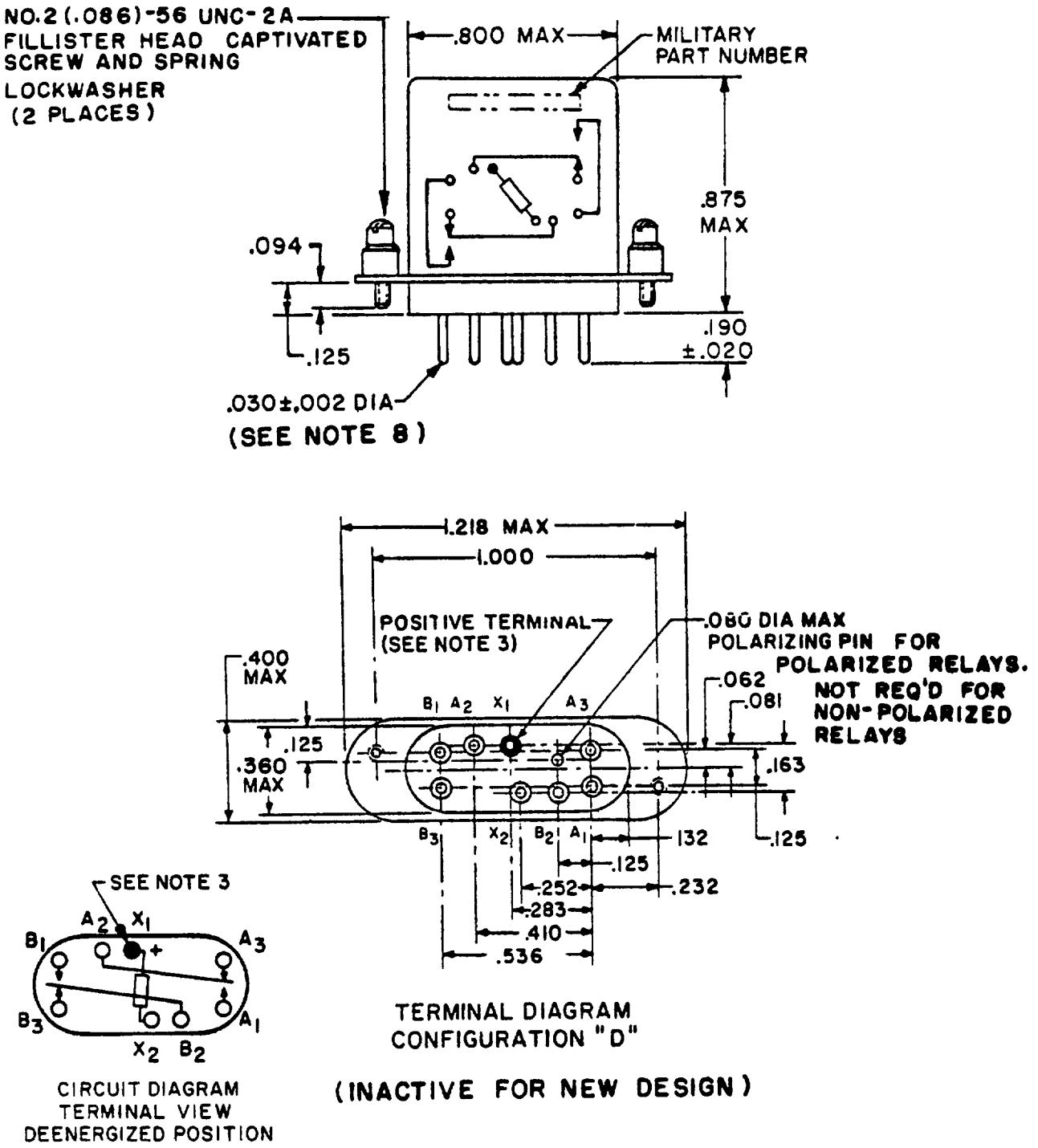



FIGURE 1. Dimensions and configuration - Continued.

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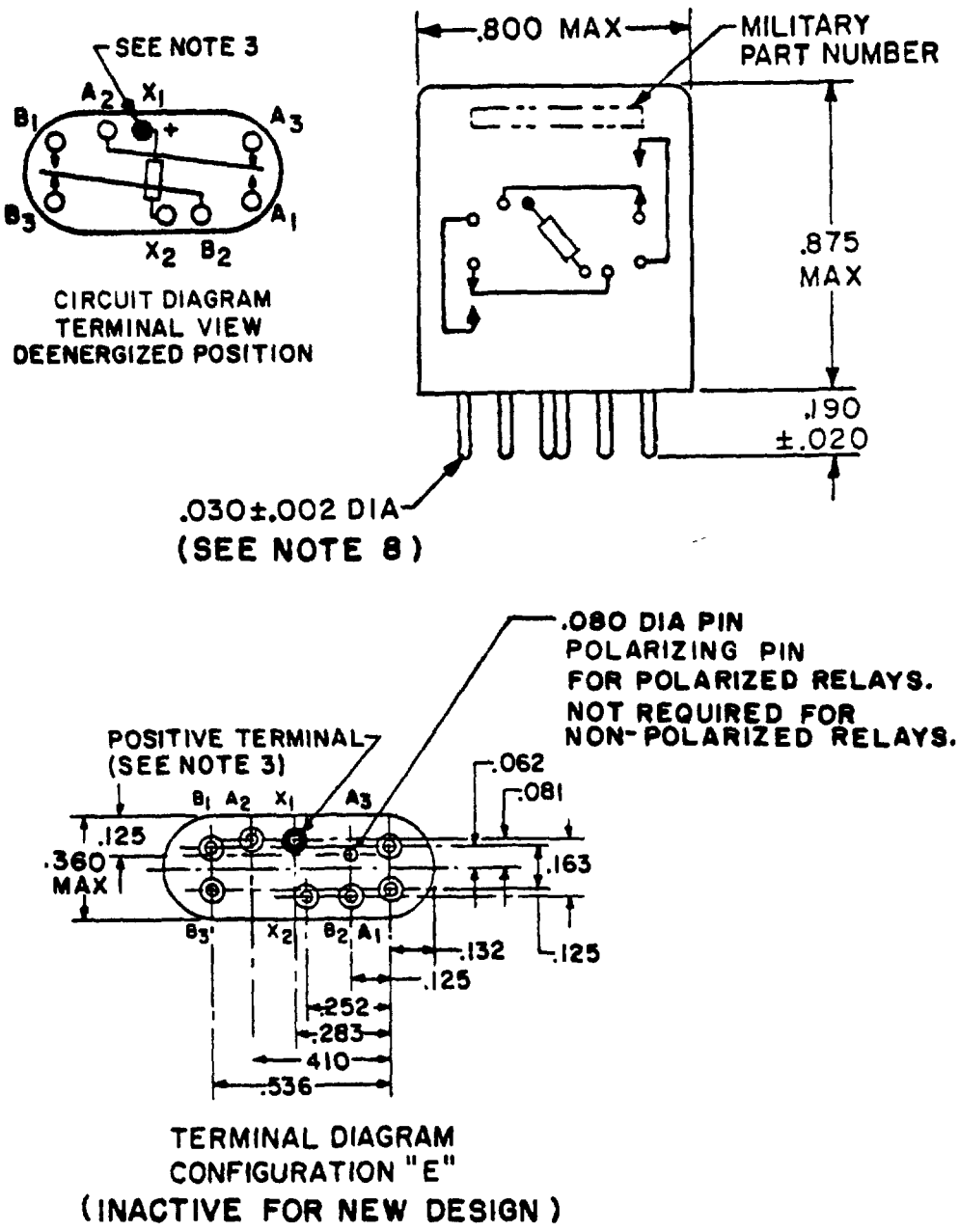



FIGURE 1. Dimensions and configuration - Continued.



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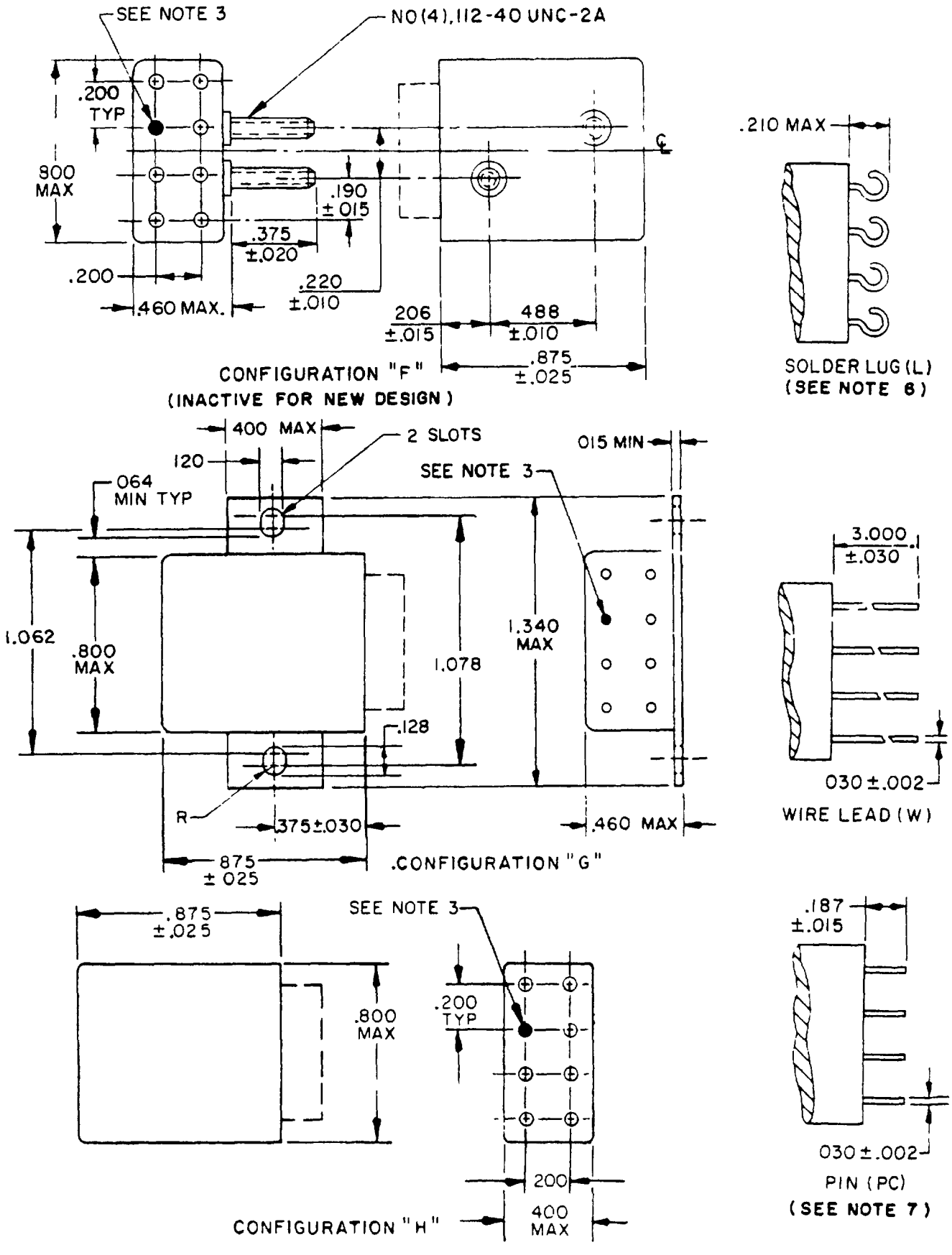
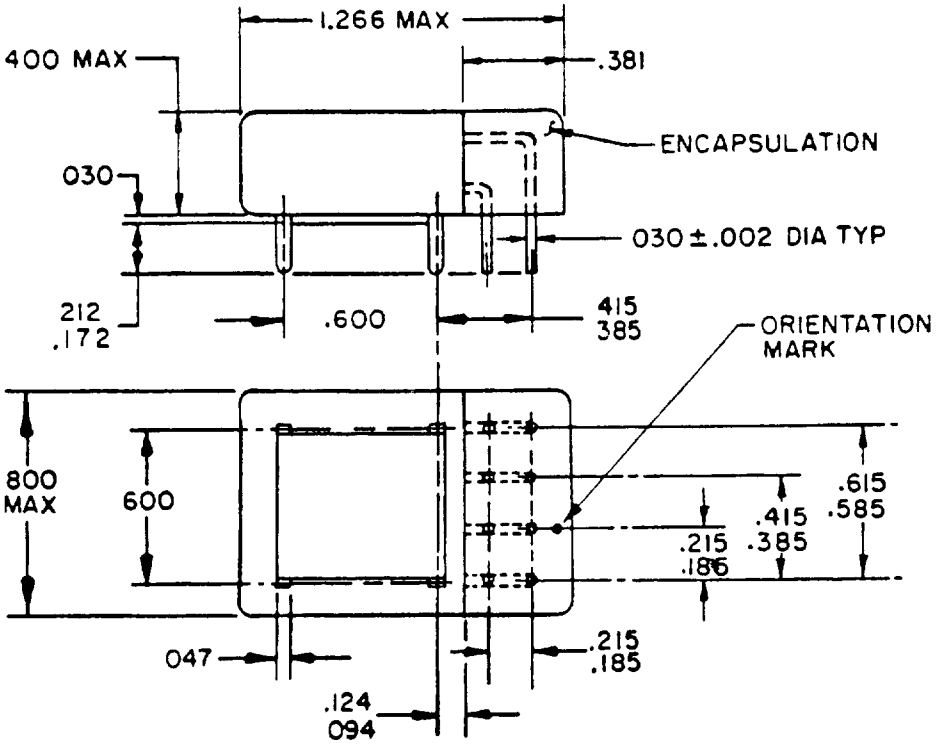



FIGURE 1. Dimensions and configuration - Continued.



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CONFIGURATION "J"

| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|---|---|
| 002 | .05 | 094 | 2 39 | 190 | 4 83 | 381 | 9 68 | 800 | 20 32 |
| 005 | .13 | .096 | 2 44 | 200 | 5 08 | 385 | 9.78 | 875 | 22 22 |
| 010 | .25 | 120 | 3.05 | .206 | 5 23 | 400 | 10 16 | 1.000 | 25 40 |
| 015 | 38 | 124 | 3.15 | 210 | 5 33 | 410 | 10.41 | 1 057 | 26 85 |
| 020 | 51 | 125 | 3.18 | 212 | 5 38 | 415 | 10 54 | 1 062 | 26 97 |
| 025 | .64 | 128 | 3 25 | 215 | 5 46 | 460 | 11 68 | 1 067 | 27 10 |
| 030 | 76 | 132 | 3 35 | .220 | 5.59 | 488 | 12 40 | 1 078 | 27 38 |
| 047 | 1 19 | 136 | 3 45 | 232 | 5 89 | 536 | 13.61 | 1 180 | 29 97 |
| 062 | 1 57 | 163 | 4.14 | 252 | 6.40 | .585 | 14 86 | 1 218 | 30.94 |
| 064 | 1 63 | 172 | 4 37 | 283 | 7 19 | 600 | 15 24 | 1 266 | 32 16 |
| 080 | 2 03 | 185 | 4 70 | 359 | 9 12 | 615 | 15 62 | 1 340 | 34 04 |
| 081 | 2 06 | 187 | 4 75 | 360 | 9 14 | 797 | 20 24 | 3 000 | 76 20 |
| | | | | 375 | 9 52 | | | | |

NOTES.
1. Dimensions are in inches. Unless otherwise specified tolerance is ±.010 (0.25 mm).
2. Metric equivalents are given for general information only.
3. Terminal X1 shall be identified by a bead of contrasting color. For tab mount orientation mark shall be used. Terminal identification for reference only.
4. Two additional mounting holes positioned as shown are optional.
5. Symmetry of terminals and mounting means with respect to the relay shall be maintained (centerline).
6. Shape of lug (solder) terminals optional.
7. Relays with pin terminals shall not be used for plug-in socket applications.
8. (K) All active electrical terminals shall be gold plated per MIL-G-45204, type II, class 1, .00005 inch (50 microinches) minimum. Gold plating of index pins is optional. Underplating, nickel 50 to 150 microinches.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

CONTACT DATA:

| | High-level characteristics 1/ | Low-level characteristics 2/ |
|---|---|---|
| Configuration | DPDT | DPDT |
| Arrangement | 2 form C | 2 form C |
| Load ratings: (Relay case grounded) | | |
| Resistive | See table I | 10 to 50 microamperes at 10 to 50 mV, dc or peak ac |
| Inductive | See table I | Not applicable |
| Lamp | See table I | Not applicable |
| Intermediate current (minimum current) | Applicable | Not applicable |
| Contact resistance or voltage drop: | | |
| Rated life: | | |
| Before | .05 ohm max | .05 ohm max |
| During | 10 percent of open circuit voltage | 50 ohms max |
| After | .10 ohm max | .15 ohm max |
| Intermediate current (minimum current): | | |
| Before | .05 ohm max | Not applicable |
| During | 3 ohms max | Not applicable |
| After | 3 ohms max | Not applicable |
| Contact bounce | 1 millisecond (ms) max | 2 ms max |
| Overload test: | | |
| Resistive | 2 times rated current | Not applicable |
| Inductive | 2 times rated current | Not applicable |

TABLE I. High-level load ratings.

| Type of load | Amperes | | |
|---|---|---|---|
| | 28 V dc | 115 V 400 Hz 3/ | 115 V 60 Hz |
| Resistive | 2.0 | 0.3 | 0.3 |
| Inductive | 1.0 4/ | 0.3 | 0.3 |
| Lamp 3/ | 0.1 | 0.1 | 0.1 |

(K) COIL DATA. See table II.

Duty rating Continuous.

Operate time (over temperature range): 6 milliseconds maximum.

Release time (over temperature range) 6 milliseconds maximum.

1/ High-level relays are not designed for use at low-level loads.

2/ Relays intended for low-level applications shall not be tested at high level or with lamps.

3/ Life test not required.

4/ At an inductive load of 200 millihenrys minimum.



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Ⓚ TABLE II. Dash numbers and applicable characteristics.

| Dash number | Configuration | Vibration (grade) | Coil voltage 1/ (V dc) (Max) | Coil voltage 1/ (V dc) (Rated) | Over temperature range: Pickup voltage (V dc) Max | Over temperature range: Hold voltage (V dc) | Over temperature range: Dropout voltage (V dc) | Coil resistance at 25°C (Ohms) | Terminals |
|---|---|---|---|---|---|---|---|---|---|
| **LOW LEVEL RELAYS** | | | | | | | | | |
| 015 2/ | F | 4 | 32 | 26.5 | 18 | 14 | 1.5 | 540 min to 770 max | L |
| 016 2/ | F | " | " | " | " | " | " | 540 min to 770 max | W |
| 022 2/ | B | " | " | " | " | " | " | 540 min to 770 max | PC |
| 064 2/ | D | " | " | " | " | " | " | 495 min to 770 max | Plug-in |
| 065 2/ | E | " | " | " | " | " | " | 495 min to 770 max | Plug-in |
| **HIGH LEVEL RELAYS** | | | | | | | | | |
| 035 | G | 4 | 32 | 26.5 | 18 | 14 | 1.5 | 540 min to 770 max | L |
| 036 | G | " | " | " | " | " | " | " | W |
| 037 2/ | F | " | " | " | " | " | " | " | L |
| 038 2/ | F | " | " | " | " | " | " | " | W |
| 039 | A | " | " | " | " | " | " | " | L |
| 040 | A | " | " | " | " | " | " | " | PC |
| 043 2/ | B | " | " | " | " | " | " | " | L |
| 044 2/ | B | " | " | " | " | " | " | " | PC |
| 052 | H | 3 | " | " | " | " | " | " | L |
| 053 | H | " | " | " | " | " | " | " | W |
| 054 | H | " | " | " | " | " | " | " | PC |
| 056 | J | " | " | " | " | " | " | 675 ±10% | PC |
| 059 | H | " | 7.3 | 6.0 | 4.2 | 3.0 | .14 | 44 ±10% | PC |
| 060 | H | " | 14.7 | 12.0 | 8.9 | 6.0 | .32 | 194 ±10% | PC |
| 066 2/ | C | 4 | 30 | --- | 3/ | 4/ | 5/ | 1000 ±5% | Plug-in |
| 067 | A | " | 14.7 | 12.0 | 8.9 | 6.0 | .32 | 194 ±10% | Plug-in |

1/ CAUTION The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
2/ Inactive for new design.
3/ Pickup current: 6.0 mA ±.5 mA dc.
4/ Hold current 5.8 mA dc.
5/ Dropout current 3.45 mA dc.

ELECTRICAL DATA

Run-in Applicable.

Insulation resistance: 1,000 megohms minimum except the resistance between coil and case at high temperature shall be 500 megohms or greater

Dielectric withstanding voltage

| | Sea level V rms (60 Hz) | Altitude V rms (60 Hz) |
|---|---|---|
| Between case, frame, or enclosure and all contacts both in the energized and deenergized position | 1,000 | 350 All terminals to case |
| Between case, frame, or enclosure and coil | 500 | |
| Between all contacts and coil | 1,000 | |
| Between open contacts in the energized and unenergized positions | 500 | |
| Between contact poles | 1,000 | |

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ENVIRONMENTAL DATA:

Temperature range· -65°C to +125°C.

Vibration: See table II.

Acceleration: Applicable.

Shock: Symbol 3 (100 G).

Resistance to soldering heat· Applicable to pin type (printed circuit) terminals.

Internal moisture: Applicable.

PHYSICAL:

Enclosure design: Symbol 4 (hermetically sealed).

Terminal strength.

Solder lug 3 pound-pull ±0.3.

Pin type. 2 pound-pull ±0.2.

Terminal twist test Applicable to wire leads.

Sealed by welding· Applicable, except configurations D and E may be solder sealed.

Dimensions and configuration See figure 1.

Termination See table II.

Weight: .050 pound (22.68 grams).

Magnetic interference· Applicable.

LIFE TEST REQUIREMENTS·

High-level: 100,000 operations per relay.

Two relays per contact rating shall be tested with rated loads on all contacts, except minimum current.

For qualification inspection, the resistive and inductive 28 V dc loads and the resistive, inductive and lamp 115 V, 60 Hz loads shall be used when performing the high-level life test.

Low-level: 100,000 operations per relay.

Intermediate current (minimum current) Level II (50, 000 operations).

QUALITY ASSURANCE:

Group A

Subgroup 1: Run-in (high level) test load 100 mA at 28 V dc utilizing 10 percent of line drop; (low level) test load 10 mA maximum at 30 mV dc maximum.



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QUALIFICATION See tables III and IV.

Ⓚ TABLE III. Qualification inspection and sample size.

| Single submission | Group submission | |
|---|---|---|
| High level<br>w/min current<br>22 units plus 1 open unit. | M5757/10-039 | 20 units plus 1 open unit.<br>Qualification inspection<br>as applicable. |
| Qualification inspection as applicable. | M5757/10-036<br>M5757/10-056 | 2 units each part number<br>Qualification inspection<br>Group I plus shock, vibration, terminal strength and seal |
| Low level<br>12 units plus 1 open unit.<br>Qualification inspection as applicable. | M5757/10-022 | 2 units<br>Qualification inspection<br>Group I plus life and seal. |

Ⓚ TABLE IV. Qualification inspection and sample size for "S" headers (configurations D and E).

| Single submission | Group submission | |
|---|---|---|
| High level<br>w/min current<br>20 units plus 1 open unit.<br>Qualification inspection as applicable. | M5757/10-066 | 20 units plus 1 open unit.<br>Qualification inspection<br>as applicable. |
| Low level<br>12 units plus 1 open unit.<br>Qualification inspection as applicable. 1/ | M5757/10-064 | 12 units plus 1 open unit.<br>Qualification inspection<br>as applicable. 1/ |
| | M5757/10-065 | 2 units, group I. |

1/ Tests as specified in the qualification inspection table of MIL-R-5757 (group V not applicable).

Part number: M5757/10- (dash number from table II).



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SUPERSESSION DATA: See table V.

Ⓚ TABLE V. Supersession data. 1/

| Superseded dash number M5757/10- | Superseding part number |
|---|---|
| 001, 007, 013 | M39016/6-107 or -207 2/ |
| 002, 008, 014 | M39016/6-108 or -208 2/ |
| 003, 009 | M5757/10-015 |
| 004, 010 | M5757/10-016 |
| 005, 011, 017 | M39016/6-104 or -204 2/ |
| 006, 012, 018 | M39016/6-105 or -205 2/ |
| 019, 021 | M39016/6-128 or -228 2/ |
| 020 | M5757/10-022 |
| 023, 029 | M5757/10-035 |
| 024, 030 | M5757/10-036 |
| 025, 031 | M5757/10-037 |
| 026, 032 | M5757/10-038 |
| 027, 033 | M5757/10-039 |
| 028, 034 | M5757/10-040 |
| 151, 051, 141, 041 | M5757/10-043 |
| 042 | M5757/10-044 |
| 061 | M39016/6-123 or -223 2/ |
| 062 | M39016/6-109 or -209 2/ |
| 063 | M5757/10-065 |
| FSCM 80063 part numbers | |
| SM-C-344272 | M5757/10-065 |
| SM-C-414133 | M5757/10-064 |
| SM-C-413824 | M5757/10-065 |
| SM-C-413823 | M5757/10-066 |

1/ Dash numbers 045 through 050, 055, 057, and 058 canceled without replacement.
2/ For Government logistics support the -100 series shall be used, for original equipment manufacturers, the -200 series may be used.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
[illegible] - MI
[illegible] - OS
Air Force - 11, 17, 99
DLA - ES
NSA - NS

User activities:
Army - AV, ME, AT
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5945-0676)

1984-505-038/A5348



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-R-5757/10K | RELAYS, ELECTROMAGNETIC, HERMETICALLY SEALED |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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